The examination is being carried out on the **following application documents**

Description, Pages

1-23 filed with entry into the regional phase before the EPO

Claims, Numbers

1-13 filed with entry into the regional phase before the EPO

Drawings, Sheets

1/17-17/17 filed with entry into the regional phase before the EPO

Reference is made to the following documents; the numbering will be adhered to in the rest of the procedure.

D1 US 2009/232462 A1 (CREEDEN DANIEL [US] ET AL) 17 September 2009

D2 US 2011/075690 A1 (CREEDEN DANIEL [US] ET AL) 31 March 2011

D3 US 2014/264024 A1 (NAKAYAMA HITOSHI [JP]) 18 September 2014

D4 US 6 144 679 A (HERMAN GREGORY S [US] ET AL) 7 November 2000

D5 US 2018/100844 A1 (HILSCHER ELKE [DE] ET AL) 12 April 2018

D6 US 2015/177155 A1 (HARAN FRANK MARTIN [CA]) 25 June 2015

1 Article 123(2) EPC:

1.1 **Claim 1** is a generalisation of original claims 14+3 which is not acceptable.
Original claim 14 or p.23, third paragraph is restricted to the plurality of excitation light sources being fibre lasers in the context of difference frequency generation method using nonlinear crystals.
All detailed embodiments are disclosing fibre lasers. Also paragraph [0055] which uses the expression "excitation light " also discloses that this expression apply to different fibre lasers.
Paragraphs [0116]-[0118] allow a generalisation of the method used outside a difference frequency generation, such as using a plurality of THz lasers or

photoconductive antenna. But when the difference frequency generation method is selected as it is the case in amended claim 1, there is no disclosure indicating that the respective excitation light sources may be something else than fibre lasers. The only generalisation disclosed is to use pulsed light instead of continuous light.

1.2 The same objection applies for **claims 2-6, 9, 11, 13**.

2 Article 52(1) EPC:

2.1 The subject-matter of **independent claim 1** lacks novelty (Article 54 EPC) because document D1 contains all the technical features defined in said claim, i.e. D1 discloses a terahertz light source (see Fig.1) comprising:
(a) a plurality of non-linear optical crystals (see Fig.2); and
(b) a plurality of excitation light sources which can be fibre lasers (see paragraph [0023]),
wherein the terahertz light source generates a plurality of terahertz waves having different wavelengths, in a same number as a number of combinations when selecting two types of excitation light sources from among the plurality of excitation light sources, by a difference frequency generation method based on angular phase matching using the non-linear optical crystal, and irradiates an object with the plurality of terahertz waves at one time (this is the case in the embodiment disclosed in paragraph [0038]).

2.2 Similarly as D1, D2 discloses the same features as defined in claim 1 (see the figures and paragraphs [0017]-[0030]).

2.3 D3 discloses also all the technical features defined in claim 1, i.e. D1discloses a terahertz light source (see Figs.1-2) comprising:
(a) a plurality of non-linear optical crystals (see Fig.2 and paragraph [0063] indicating that the light source may be obtained with a difference frequency generation system implying the use of non-linear crystal for each THz source 22); and
a plurality of excitation light sources (see Fig.2 and paragraph [0063]. NB: the use of fibre lasers as light sources in a difference frequency generation method does not involve any inventive activity),
wherein the terahertz light source generates a plurality of terahertz waves having different wavelengths (this is implied by Fig.6 and paragraphs [0059],

[0083]-[0090]), in a same number as a number of combinations when selecting two types of excitation light sources from among the plurality of excitation light sources, by a difference frequency generation method based on angular phase matching using the non-linear optical crystal, and irradiates an object with the plurality of terahertz waves at one time (see Figs.1-2, 6-7 and paragraphs [0059], [0063], [0083]-[0090]).

2.4 The subject-matter of **claims 2-13** is not novel in view of D1, D2 and/or D3 (Art. 54 EPC) or does not involve an inventive step (Art.56 EPC) in the context of these documents:
Claims 2-5: see Fig.2 and paragraphs [0017]-[0030] in D2. As explained in paragraph [0020] of D2, the skilled person will adapt the illumination conditions according to phase matching conditions associated with the non-linear crystals and the wavelengths of the light sources. This implies a control of the angular conditions. Also, starting from D1 or D3, the arrangement defined in said claims does not appear to involve an inventive, in particular, depending of the required wavelengths for the pump and signal in order to obtain the desired THz wavelengths, the use of beam splitters (such as polarisation beam splitters) to use several times the same pump or signal for the different THz sources is considered to be obvious. The use of optical fibre to guide the pump and signal light beams is also an obvious design option;
Claims 6, 8-10, 13: see Figs.1-2, 6-7 in D3; the application of a multi-wavelength THz light source as taught in D1 and D2 to inspection and identification of materials is suggested in D1 and D2 (see paragraph [0003] in D1 and paragraph [0002] in D2);
Claim 7: GaP is a well-known nonlinear crystal for THz light generation using the difference frequency generation method (see D4, the abstract and Figs.1,3);
Claim 11: depending on the sample to investigate, pressure means are known to be used to provide the sample in a suitable arrangement for THz; measurements (see Fig.1 and paragraphs [0032], [0038] in D5);
Claim 12: the claimed arrangement is considered to be a known alternative for flat pieces to be inspected (see Fig.9 and paragraphs [0028], [0041]).

3 Other remarks:

3.1 To meet the requirements of Rule 42(1)(b) EPC, the documents D1-D3 should be identified in the description and the relevant background art disclosed therein should be briefly discussed.

3.2 The independent claims are not in the two-part form in accordance with Rule 43(1) EPC, which in the present case would be appropriate, with those features known in combination from the prior art being placed in the preamble (Rule 43(1)(a) EPC) and with the remaining features being included in the characterising part (Rule 43(1)(b) EPC).

3.3 The features of the claims should be provided with reference signs placed in parentheses to increase the intelligibility of the claims (Rule 43(7) EPC). This applies to both the preamble and characterising portion.

3.4 The word 'spirit' in the description in paragraph [0120] as well as the claim-like aspects in paragraph [0120] implys that the subject-matter for which protection is sought may be different to that defined by the claims, thereby resulting in lack of clarity of the claims (Article 84 EPC) when used to interpret them (see the Guidelines, F-IV.4.4). These statements should therefore be amended to remove this inconsistency.
Also the expression "incorporated by reference" in paragraph [0001] should be deleted (see Guidelines F-III.8).

3.5 When filing amended claims the applicant should at the same time bring the description into conformity with the amended claims. Care should be taken during revision, especially of the introductory portion and any statements of problem or advantage, not to add subject-matter which extends beyond the content of the application as originally filed (Article 123(2) EPC).

3.6 In order to facilitate the examination of the conformity of the amended application with the requirements of Article 123(2) EPC, the applicant is requested to clearly identify the amendments carried out, irrespective of whether they concern amendments by addition, replacement or deletion, and to indicate the passages of the application as filed on which these amendments are based.